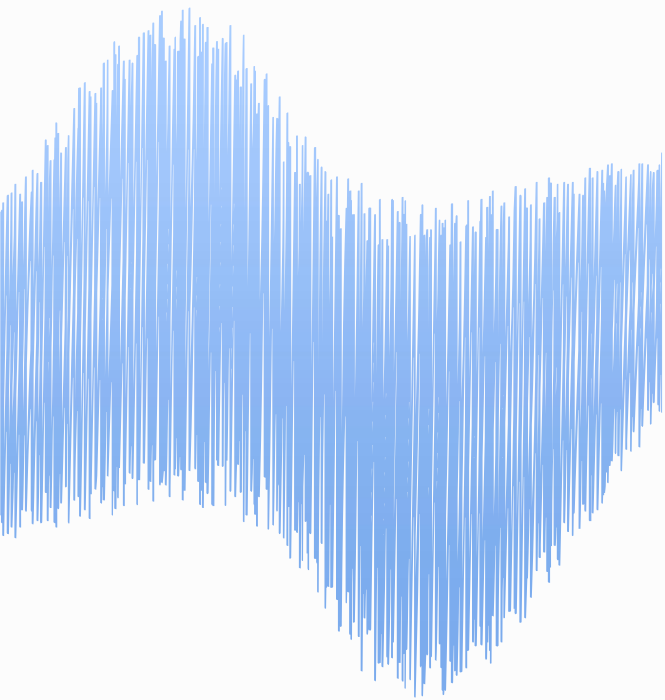




Fake vs Real: Deepfake Audio Detection

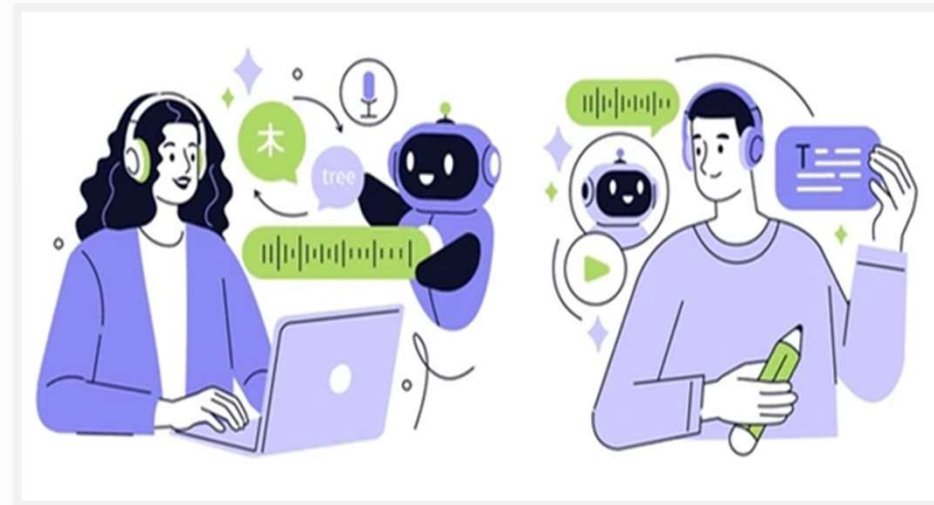
Ashley S. Nguyen Kimberly P. Nguyen Taylen Washington

Background and Motivation

Artificial intelligence has made it possible to generate highly realistic human like voices using text-to-speech technologies. While these advancements are useful for applications such as virtual assistants and accessibility tools, they also introduce serious risks.

Synthetic speech can be used for impersonation, fraud, and the spread of misinformation, making it difficult to determine whether an audio recording is real or fake.

The goal of this project is to develop machine learning models that can automatically classify audio recordings as either real human speech or AI-generated speech.



Dataset Overview



Dataset Info

- Fake-or-Real (FoR) Audio Dataset (Kaggle)
- Contains real human speech and AI-generated (fake) speech



Problem Type

- Binary classification problem
- Real vs Fake



Data Structure

- Organized into folders:
 - Real
 - Fake
- Audio format: .wav



Data Summary

- Dataset is fairly balanced
- Average duration:
 - Real: ~5.39 sec
 - Fake: ~2.36 sec
 - Difference: ~3.03 sec

Technical Challenges: **Audio Duration Bias**

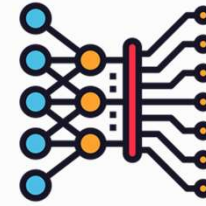
- Real audio is ~3 seconds longer on average
- Risk: model may learn duration instead of speech patterns
- Solution: convert audio to fixed-size features
- Used MFCC + fixed-size mel-spectrograms
- It's important that the model focuses on patterns, not length

Method Overview



Traditional Machine Learning

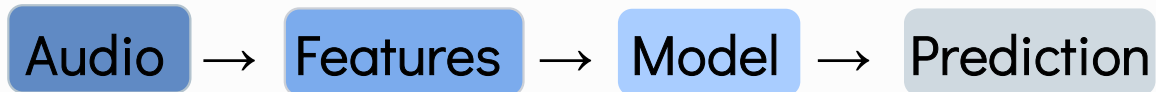
- MFCC Features (Mean + Std)
- Random Forest
- Support Vector Machine (SVM)
- Logistic Regression



Deep Learning

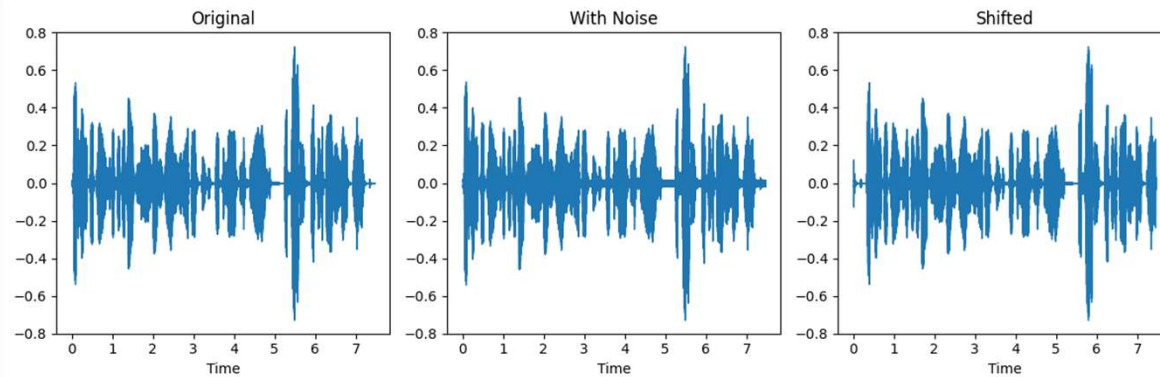
- Mel Spectrograms
- CNN Model
- Learns patterns from audio

Pipeline



Data Augmentation

Applied data augmentation by adding noise and shifting audio to create additional training samples.



Each original audio file is transformed into multiple versions to improve model generalization.



Noise

- Adds small, random variations to the audio signal



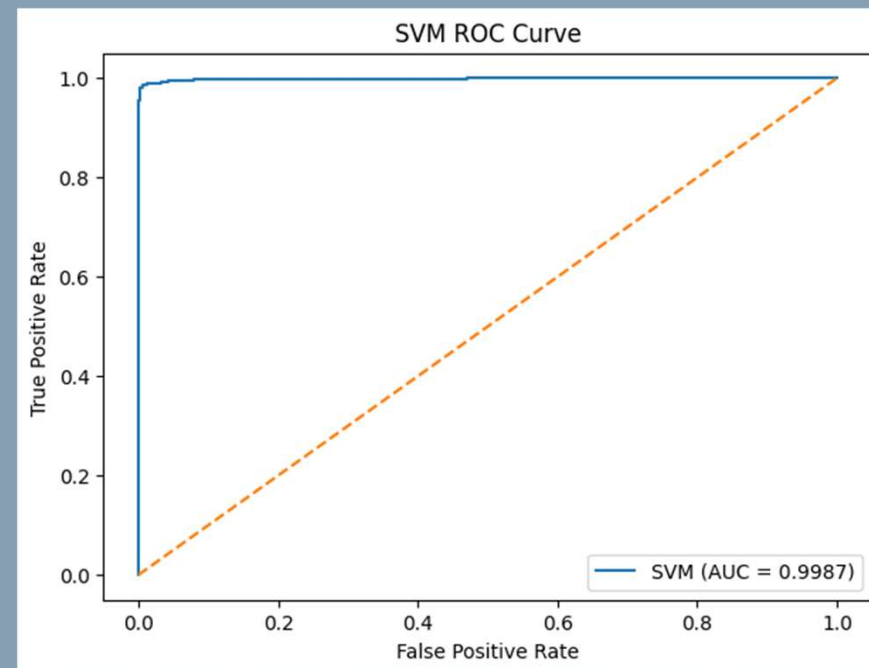
Shift

- Moves the audio slightly in time, preserving the speech

These subtle changes help CNN learn more general patterns and improve performance on unseen data.

Traditional ML Results

	Accuracy:	F1:
Logistic Regression:	~91.49%	~91.41%
Random Forest:	~97.74%	~97.73%
SVM:	~92.54%	~92.33%
Logistic Regression (Scaled):	~91.54%	~91.46%
SVM (Scaled):	~98.84%	~98.84%



Feature scaling significantly improved SVM performance, making it the best model.

Feature Importance

What we did

Extracted MFCC features from audio.

We used:

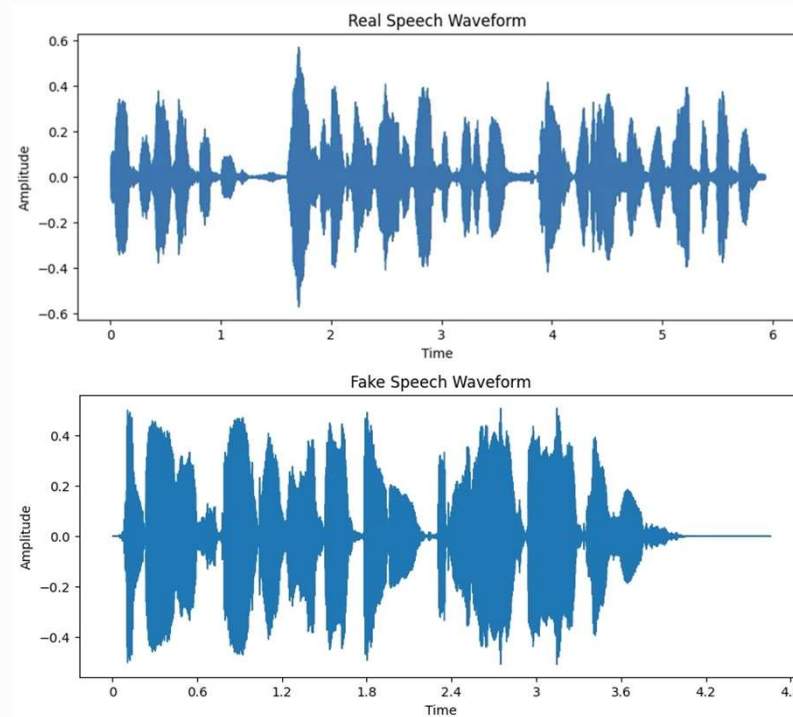
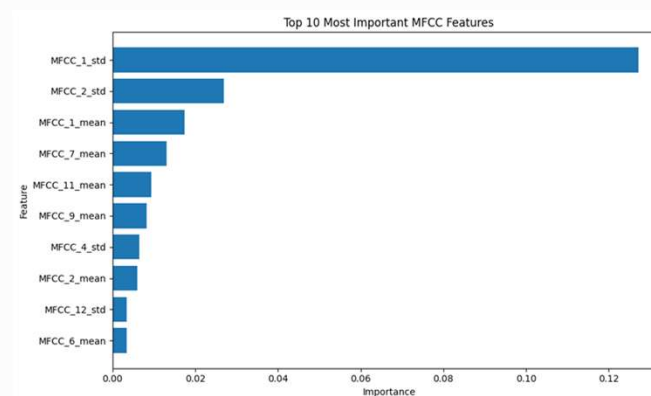
- Mean: The average frequency content.
- Standard Deviation: The variation over time.

Key Insight

- Most important features is MFCC standard deviation.
- Captures how audio changes over time.

Why it matters

- Real speech is more natural variation
- Fake speech is often smoother



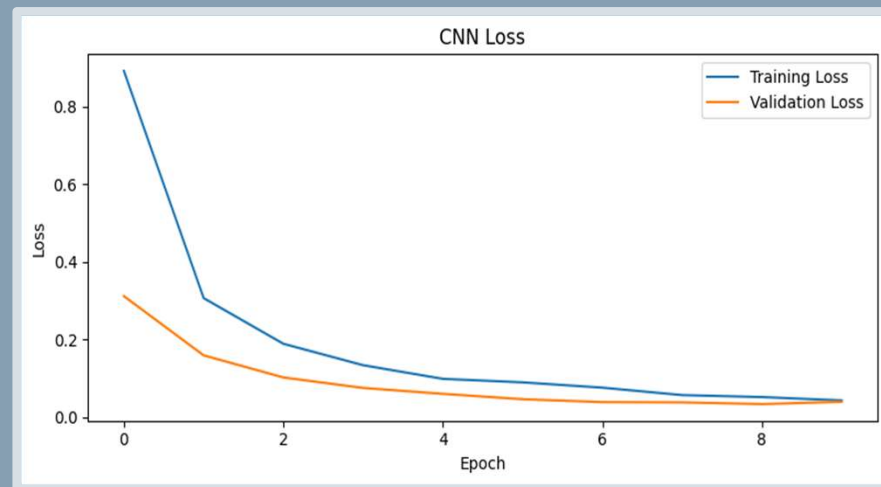
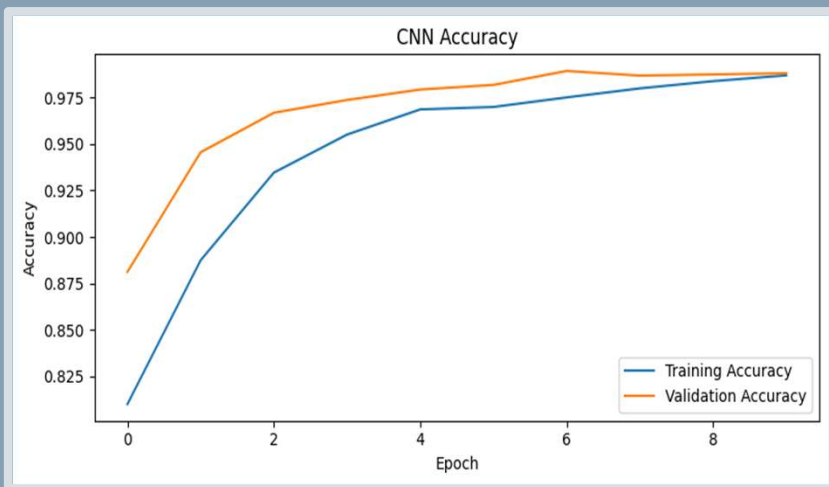
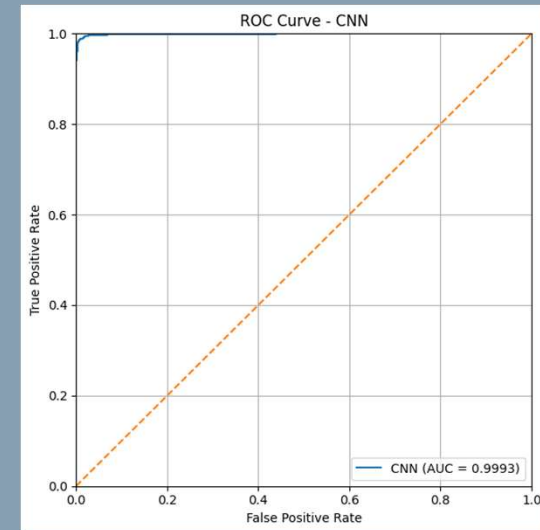
CNN Results

Training converged quickly in just a few epochs.

Accuracy: ~98.82%

F1: ~98.81%

- ROC curve evaluates classification performance
- AUC ≈ 0.9993
- Strong separation between real and fake audio



CNN's Pattern Learning

What CNN does

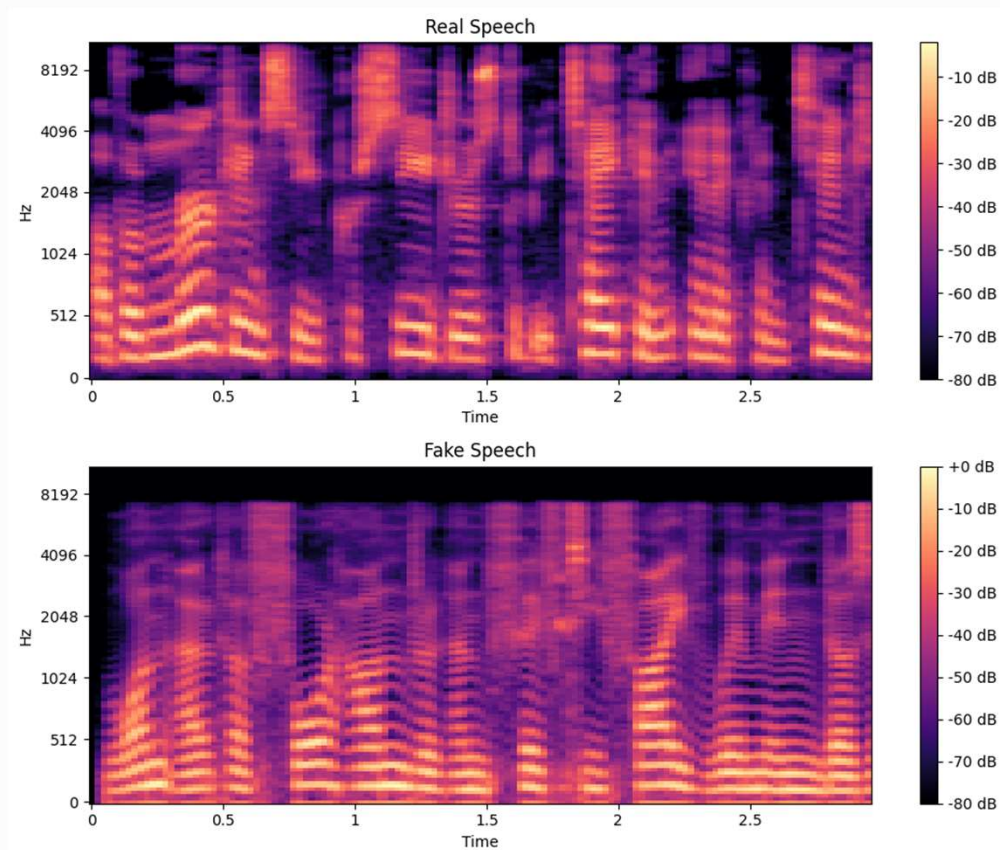
- Converts audio to Mel spectrograms (2D images)
- Learns patterns in spectrograms
- Automatically detects real vs fake speech

Key Insight

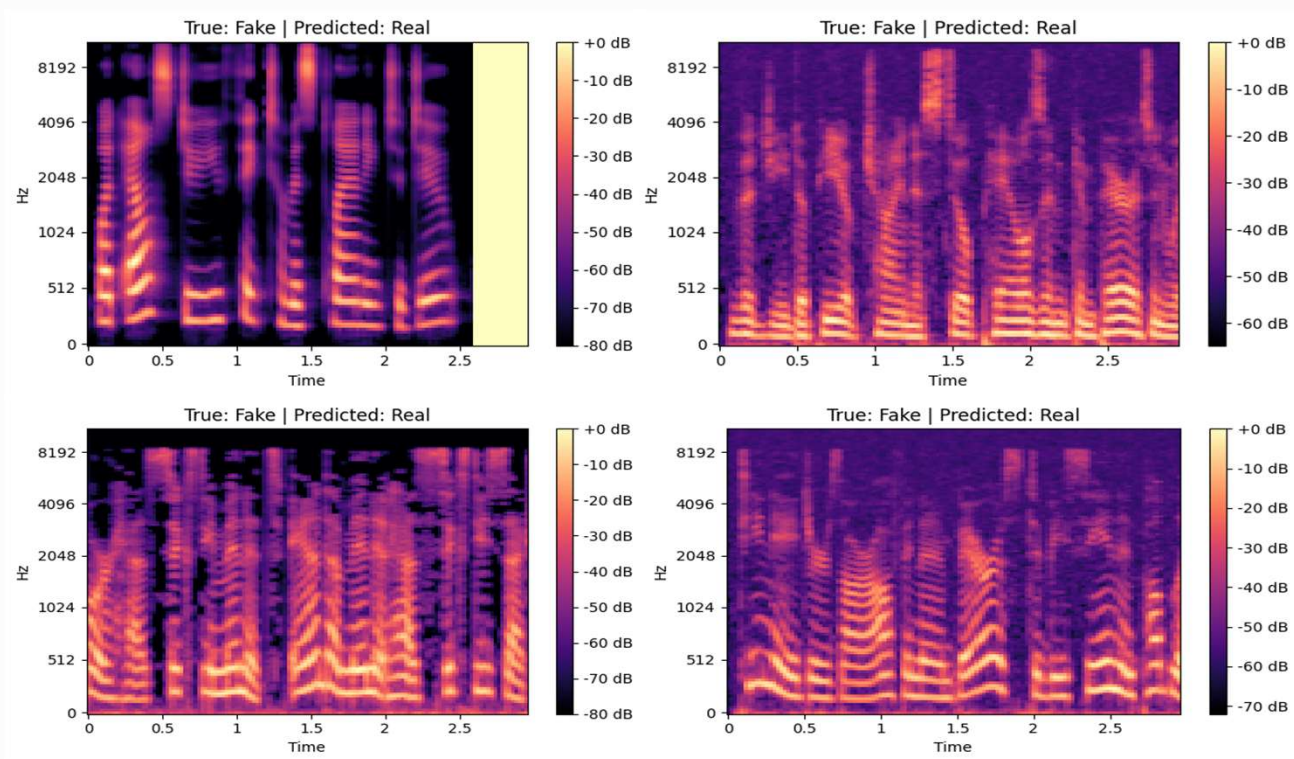
- CNN focuses on unique visual differences between real and fake speech
- Trains directly on spectrogram images instead of manual features

Why it matters

- Real and Fake speech have distinct patterns in the spectrograms
- Fake speech often has smoother, less natural patterns

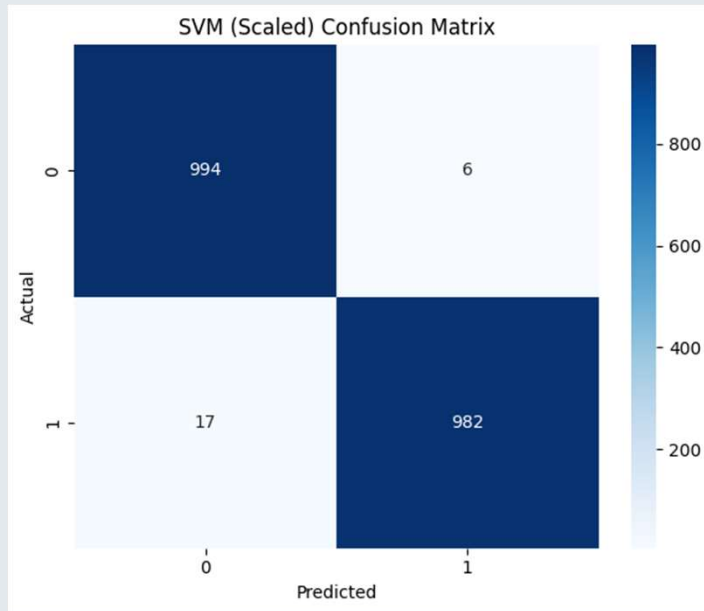


Model Errors (Misclassified Examples)

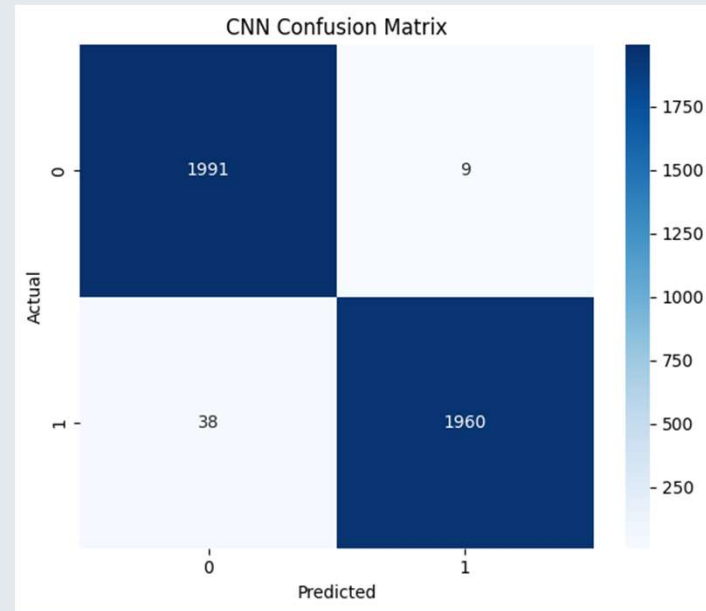


These examples show where the model struggles when real and fake patterns are similar.

Confusion Matrix Comparison



Accuracy: ~98.84%
F1: ~98.84%



Accuracy: ~98.82%
F1: ~98.81%

SVM achieved better performance by leveraging MFCC features directly, whereas CNN struggled with limited data.

Specialized Datasets

ASVSpoof19

- Less realistic, designed for lab testing
- Contains the largest variety of deepfake techniques
- 5000 samples; 2500 fake, 2500 real

In-the-Wild

- Real-world deepfakes
- Sampled from internet videos and social media
- 2364 samples: 1182 fake, 1182 real

MLAAD/ M-AILABS

- Foreign language dataset; German
- Detecting English speaker bias
- 5000 samples: 2500 fake, 2500 real

Best Results/Models for Comparison

ASVSpooof19

Spectrogram CNN

Acc: 70.82%

EER: 28.68%



- Best at catching computer-generated deep fakes
- Spots digital fingerprint features like sound quality

In-the-Wild

Spectrogram CNN

Acc: 66.67%

EER: 31.81%



- Generalizes the best on real-world data
- Most resistant to background noise

MLAAD

SVM (Unscaled)

Acc: 77.66%

EER: 12.36%



- Optimizes less features
- Robust to different languages

Next Steps

Step 1

Increase Dataset Diversity

Expand dataset with more voices, accents, and recording environments.

Step 2

Improve Model Robustness

Train the model to handle noise, shorter clips, and varying audio quality.

Step 3

Real-Time Deepfake Detection

Implement the model to detect fake audio in real time for security and fraud prevention.

Step 4

Secure Deployment

Develop controlled systems for detecting deep fake audio in security and verification settings.



Conclusion



What we did

- Built a model to detect real vs fake (deepfake) audio
- Compared multiple models:
 - Random Forest
 - Logistic Regression
 - SVM (w/ & w/o scaling), CNN (mel spectrograms)
- Evaluated the models

Key Insight

- Feature scaling significantly improved SVM performance
- MFCC features are very effective for detecting fake audio
- CNN works well, but SVM with engineered features outperformed it

Final takeaway:

- Best model: SVM (Scaled)
- Traditional ML + good features can outperform deep learning for detecting deepfake audio

Results

- SVM (Scaled) performed the best
Accuracy: ~98.8%, F1: ~98.8%
- CNN also performed very well
Accuracy: ~98.7%, F1: ~98.8%
- Other models performed slightly lower but pretty well

Practical Value

- Deepfake audio is increasing in
 - Fraud, voice impersonation,
 - Misinformation, cybersecurity
- This model shows we can accurately detect fake audio with high reliability

**Thank
You**